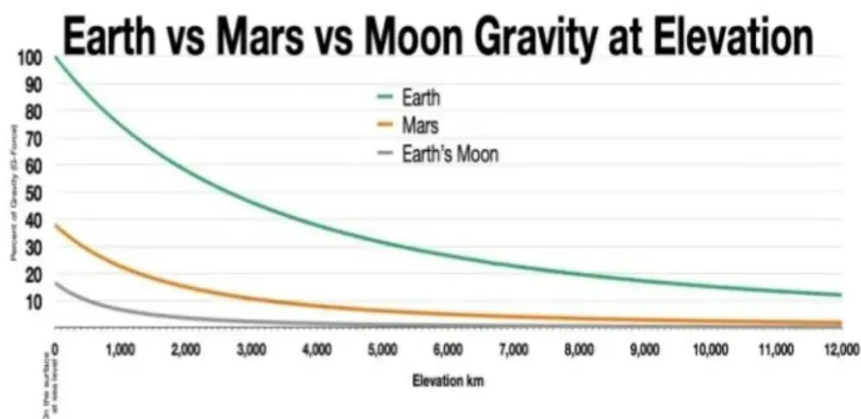


Topic:

Take WEIGHTS of five of your friends. Find out what their weights will be on the MOON AND THE MARS.

Introduction:-

Weight is the force exerted by gravity on an object, which varies depending on the gravitational pull of the celestial body the object is on. Earth's gravity is stronger than the gravity on the Moon and Mars, meaning objects weigh less on these bodies. The gravitational force on the Moon is about 1/6th of Earth's, while on Mars, it is approximately 0.38 times that of Earth. The weight of an object depends on the gravitational pull of the celestial body it is on. Earth has a standard gravity of 9.8 m/s^2 , while the Moon's gravity is about 1/6th of Earth's, at approximately 1.63 m/s^2 . Mars has a gravity of 3.71 m/s^2 , which is about 0.38 times that of Earth. This means that objects (including people) weigh less on the Moon and Mars than they do on Earth.



Earth's gravity (g) : 9.81 m/s^2

Moon's gravity : 1.62 m/s^2

Mars' gravity : 3.71 m/s^2

Weight Conversion Formula:-

The weight of an object on another celestial body can be calculated using the formula:

Weight on celestial body = Weight on Earth $\times$ (Gravity on celestial body / Gravity on Earth)

Students Data and Weight Calculation:-

Name Weight

Raj 50 kg

Shivam 55 kg

Rohit 60 kg

Sahil 65 kg

Manoj 70 kg

Full Calculations:-

Weight on the Moon:-

Weight on Moon = Weight on Earth $\times (1.62/9.8)$

Weight on Mars:-

Weight on Mars = Weight on Earth $\times (3.71/9.8)$

1. Student 1 (50 kg on Earth):

Moon:- $50 \times (1.62/9.8) = 8.32 \text{ kg}$

Mass:- $50 \times (3.71/9.8) = 18.93 \text{ kg}$

2. Student 2 (55 kg on Earth):

Moon:- $55 \times (1.62/9.8) = 9.14 \text{ kg}$

Mass:- $55 \times (3.71/9.8) = 20.82 \text{ kg}$

3. Student 3 (60 kg on Earth):

Moon:- $60 \times (1.62/9.8) = 9.98 \text{ kg}$

Mass:- $60 \times (3.71/9.8) = 22.73 \text{ kg}$

4. Student (65 kg on Earth):

Moon:- $65 \times (1.62/9.8) = 10.82 \text{ kg}$

Mars:- $65 \times (3.71/9.8) = 24.63 \text{ kg}$

5. Student 5 (70 kg on Earth):

Moon:- $70 \times (1.62/9.8) = 11.66 \text{ kg}$

Mars :- $70 \times (3.71/9.8) = 26.54 \text{ kg}$

Summary Table:-

Student Name	Weight on Earth (kg)
Student 1	50
Student 2	55
Student 3	60
Student 4	65
Student 5	70

Weight on Moon (kg)	Weight on Mars (kg)
8.32	18.93
9.14	20.82
9.98	22.73
10.82	24.63
11.66	26.54

Conclusion:-

This exercise demonstrates how gravitational differences between Earth, the Moon, and Mars affect weight. By understanding these differences, students can gain insights into fundamental concepts of gravity and mass. Calculating weights on different celestial bodies illustrates the practical application of physics in space exploration and provides a foundation for more advanced studies in astronomy and physics.